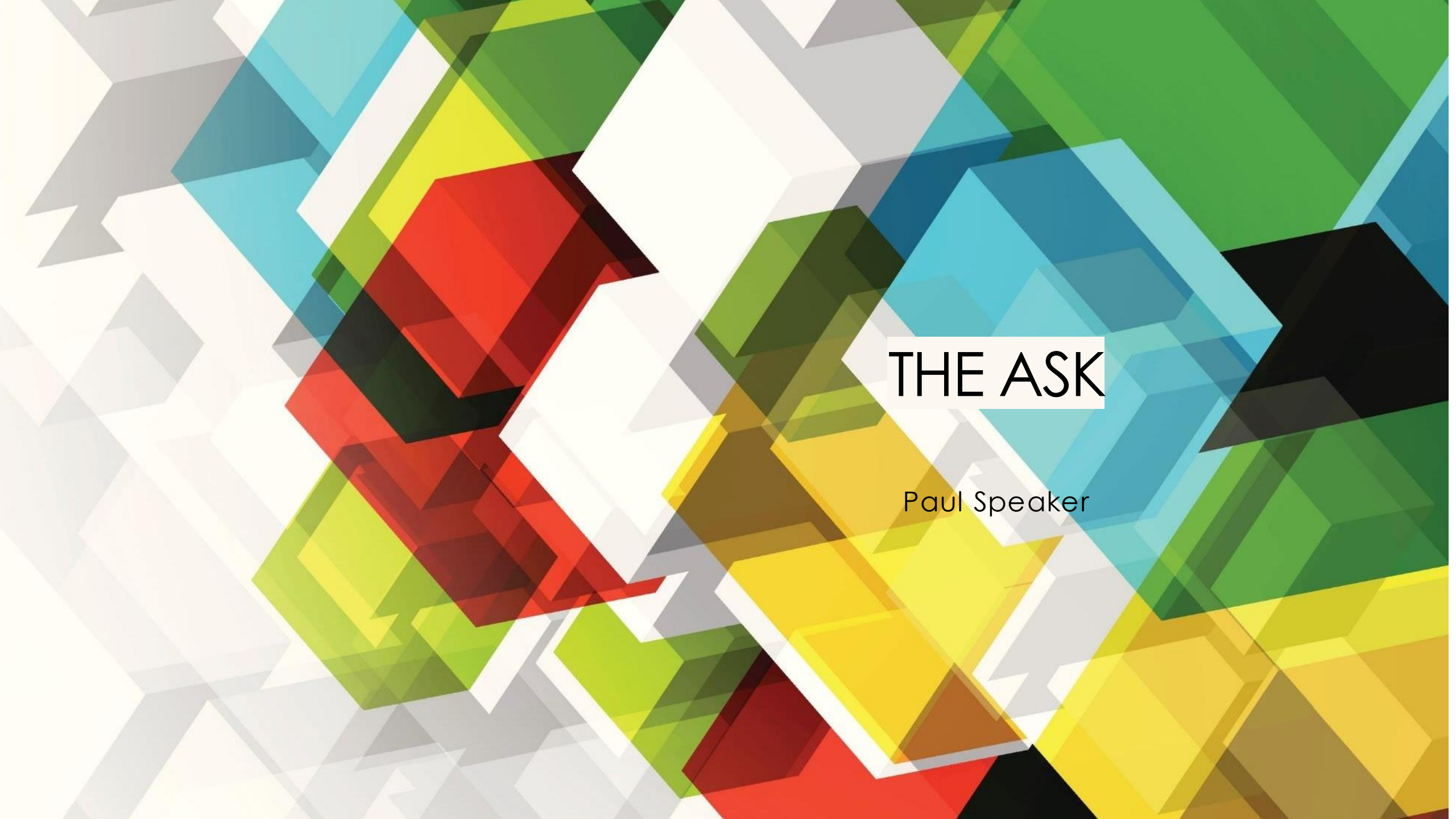




THE ASK

Paul Speaker

Where Babies (Data Science Problems) Come From

- Where do data science projects come from to get to entry level data scientist?

- Usually from a senior data scientist.
- But where do they get it from?
- The answer can vary from organization to organization
- Many “models” of what data science orgs look like

- Centralized

single data science organization that can be in a lot of different functions, often within an IT organization

every function of the organization will come to you with a question (supply chain, marketing, ...), all self-contained

- Functional alignments

each department/function has its own data scientists in-house

- Hub and spoke

centralized, but branches for each function, each branch has its own data scientists (Goldilocks)

- Data science orgs do not generate their own problems—they come from someone with a need

- Who are you talking with first?

person driving the ask may not be the one you work with

- Is it the same as the one asking (or is it someone below)?

could be hostile - don't want you to be there

answer looks different depending on who is asking (e.g., someone from finance vs. someone from sales)

Disney has 100+ data scientists who do nothing but price optimization

the larger the initiative, the higher up it comes from in the chain

The Needs

- People do not come to you with data science problems. They come to you with problems
- “We have too many delivery delays. What can we do about that?”
- “Our inventory is out of wack.”
- “Why do customers hate us?”
- “Employee expenses are way up. Are they ripping us off?”
- “Equipment failures are causing too much downtime.”
- “Why are these competitor products going viral?”
- “Which customers can we get away with charging more?”
- “<This group of employees> is spending too much time doing <task X>.”
- “Why are our best employees leaving?”

Examples
(diluted) from Dr.
Speaker's
experience

Creating a Data Science Solution

What to do with the Needs?

- For each of these, we need to figure out a prediction that would help to solve the problem
- Sometimes, the ask does not really require a prediction
 - “Which products are selling the most this month?”
 - “How many customers are buying the same product over and over again?”
 - It's okay to say so!
 - A good data science organization says no (or redirects) inappropriate projects
- How can you be sure?
 - Communication is key
 - It is shocking how often this does not happen

often, d.s. receive vague instruction, vanish for months to create elaborate solution that DOES NOTHING

continuous communication is essential (even if instructions clear)

often, data science org closely linked or partnered with reporting organization (think Emendata + DOJ)

“Data Science Solution” very vague, whatever you make it

it is also ok to answer a yes/no question with “not yet”

Creating a Data Science Solution

- To be a solution, the prediction has to create a change in behavior
 - No change in behavior → no value
- So you have to identify the behavior
- Identify the decisions being made (including automated)
- Identify the object of decision
 - Units in space and time
- What does data look like? Focus on target first
- If you knew X, what would you do differently?
- Today we are talking about the model identification. We will save other topics for later, such as
 - Constraint identification
 - Process mapping